

Lesson 10 Homework - Calculus

William Wang

September 2026

1

We have the function

$$f(x) = \sqrt{|4 - x|}$$

To see whether it is differentiable at 4 we want to see if the right hand derivative is equal to the left hand derivative. Right hand derivative

$$\begin{aligned}\lim_{h \rightarrow 0^+} \frac{f(4+h) - f(4)}{h} &= \lim_{h \rightarrow 0^+} \frac{\sqrt{|4 - 4 - h|} - \sqrt{|4|}}{h} = \lim_{h \rightarrow 0^+} \frac{\sqrt{h}}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}} = \infty\end{aligned}$$

Then the left hand derivative is

$$\lim_{h \rightarrow 0^-} \frac{f(4+h) - f(4)}{h} = \lim_{h \rightarrow 0^-} \frac{\sqrt{|h|}}{h}$$

since $h < 0$,

$$= \lim_{h \rightarrow 0^-} \frac{\sqrt{-h}}{h}$$

since $h < 0$, we can write h as $-(-h)$. Then

$$\lim_{h \rightarrow 0^-} \frac{\sqrt{-h}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{-h}}{-(\sqrt{-h})^2} = \lim_{h \rightarrow 0} \frac{1}{-\sqrt{-h}} = -\infty$$

Graphically, as x approaches 4 from the right, the tangent slopes become infinitely steep and positive. As x approaches 4 from the left, the tangent slopes become infinitely steep and negative. This forms a shape that looks like 2 semi circles tangent to each other and opening away from each other.